Python Programming Foundations

Week 7 - RBA & Project Framing

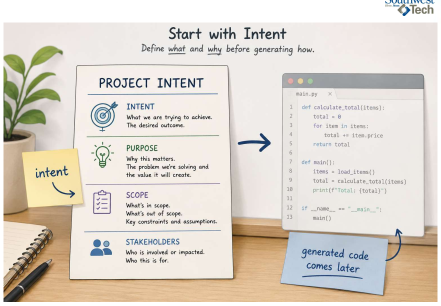
Day 1: "Intent Before Creation"

1

Intent Should Exist Before Generation

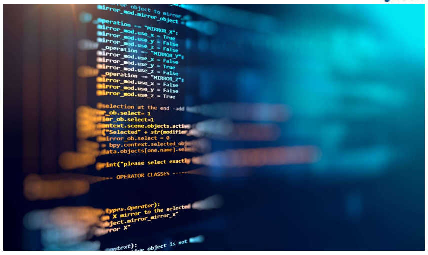
- AI can generate code quickly, but speed does not prove the project is clear.
- Before generation, we need to know what the project is for, what it should do, and what decisions should remain human decisions.

2



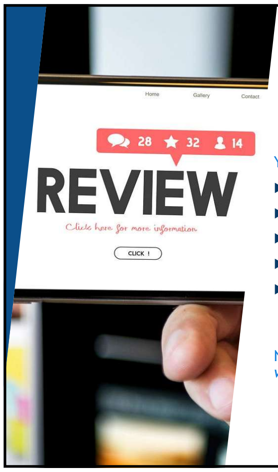
Intent Before Code

3



Review Coding Lab (Opt)

4



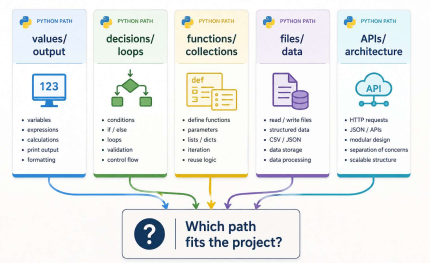
Review: You Have Several Python Paths Now

You have practiced:

- values and output
- decisions and loops
- functions and collections
- files and structured data
- APIs and architecture recognition

Now the question becomes:
which path fits the project?

5



Review: You Have Several Python Paths Now

6

What Counts as Success Today

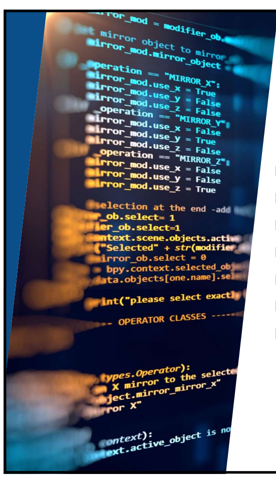
- define project purpose
- identify inputs and outputs
- name constraints and risks
- choose a likely structure
- set AI-use boundaries
- produce framing that can support capstone approval

7

Southwest Tech

Today's Success

8



Southwest Tech

What We Will Use Today

- purpose
- inputs
- outputs
- constraints
- possible structure
- risks
- AI boundaries

These are project decisions before they are coding decisions.

9

Toolbox

10

What We Will "Skip" For Now

- full build work
- polishing features
- publishing packages
- advanced dependency management
- asking AI to replace the project plan

Today is framing before building.

11

Parked For Later

12

Project Design

13

Purpose Comes First

The project purpose answers:

- What problem or need does this address?
- Who uses it?
- What would count as a useful result?

If the purpose is vague, everything after it becomes harder to judge.

14

Purpose Comes First

15

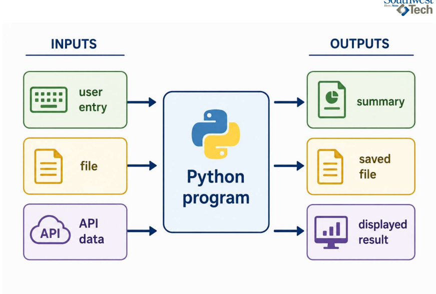
Inputs And Outputs Define The Shape

Inputs are what the program receives.

Outputs are what the program produces.

Clear inputs and outputs make the project easier to build, test, and explain.

16



Inputs & Outputs Define The Shape

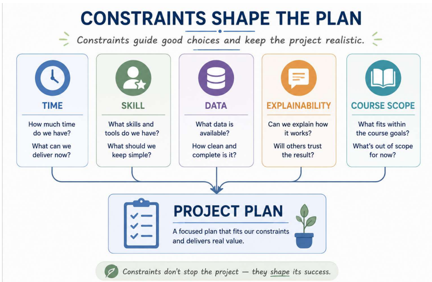
17

Constraints Are Design Information

Constraints are not just obstacles.

Time, skill level, data availability, explainability, and course scope all help shape a realistic project.

18



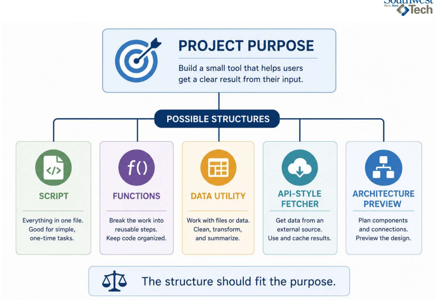
Constraints Are Design Information

19

Structure Should Fit The Intent

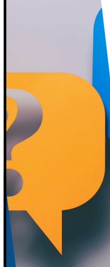
- simple script
- function-based tool
- data-driven utility
- API-style fetcher
- light architecture-preview project, if approved
- (The structure should fit the purpose.)

20



Structure Should Fit The Intent

21

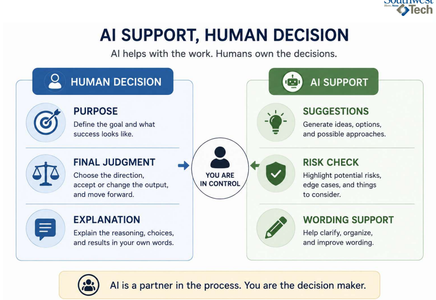


AI Boundaries Preserve Ownership

Before using AI, decide:

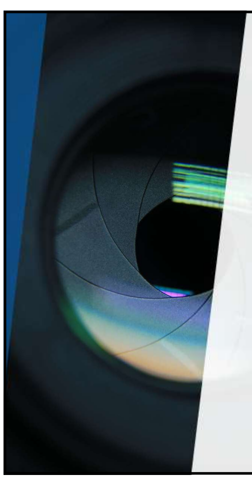
- what AI may help with
- what AI should not decide
- what you must verify
- what you must explain yourself

22



AI Boundaries Preserve Ownership

23



Demo #1

“Prompt-First vs. Intent-First”

24

Southwest Tech

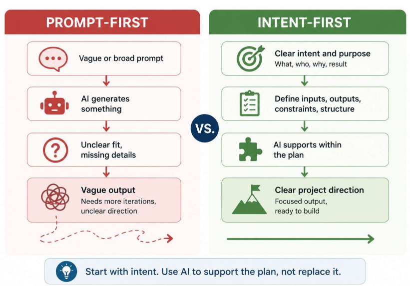
What to Watch For

Watch how the project changes when we start with intent instead of a vague prompt.

- The goal is not a prettier prompt.
- The goal is better judgment.

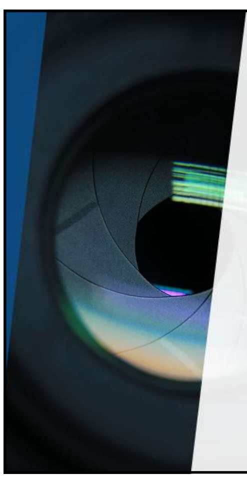
25

Southwest Tech



Demo 1: Prompt-First vs Intent-First

26



Demo #2

"Framing A Study Timer"

27

Southwest Tech

What to Watch For

A framing artifact should make the project easier to understand before code exists.

Look for:

- Purpose
- Inputs
- Outputs
- Constraints
- Structure
- AI boundaries.

28

Southwest Tech

STUDY-TIMER FRAMING WORKSHEET
Define the plan before building the timer.

Project Name: ____________
Date: ____________

1. PURPOSE
What is this study timer for?
What result should it help the user achieve?

2. INPUTS
What information or actions will the timer need?

3. OUTPUTS
What should the timer show, save, or provide?

4. CONSTRAINTS
What limits or rules should shape this project?

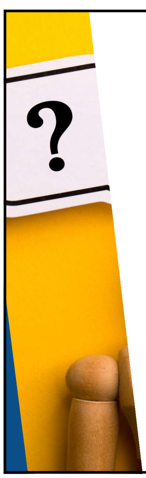
5. AI BOUNDARIES
How will AI be used as support, not replacement?

AI CAN HELP WITH	HUMAN WILL DECIDE
•	•
•	•
•	•

Use this plan to guide your build.
A clear plan today saves time tomorrow.
Next Step: ____________

Demo 2: Framing A Study Timer

29



Southwest Tech

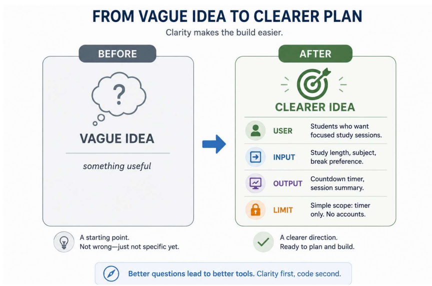
Common Failure: Vague Ambition

"I want to make something useful"

… is **not** enough.

- Useful for whom?
- With what input?
- Producing what output?
- Under what limits?

30



Common Failure: Vague Ambition

31

Assignment #13

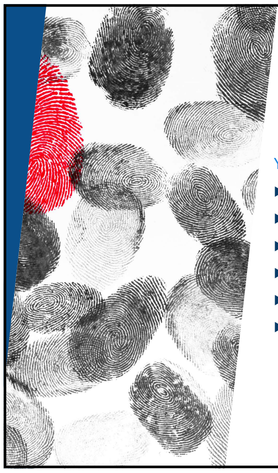
For Assignment 13, create a project framing document.

This can become the raw material for your capstone proposal, so make it concrete enough to review and revise.

32

Assignment 13

33



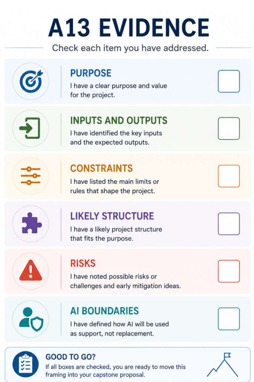
Southwest Tech

Evidence For A13

Your framing should include:

- project purpose
- inputs and outputs
- constraints
- likely structure
- risks or unknowns
- AI-use boundaries

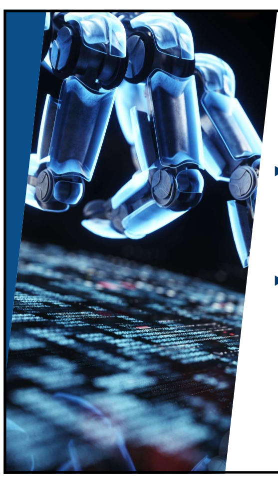
34



Southwest Tech

Evidence Example

35

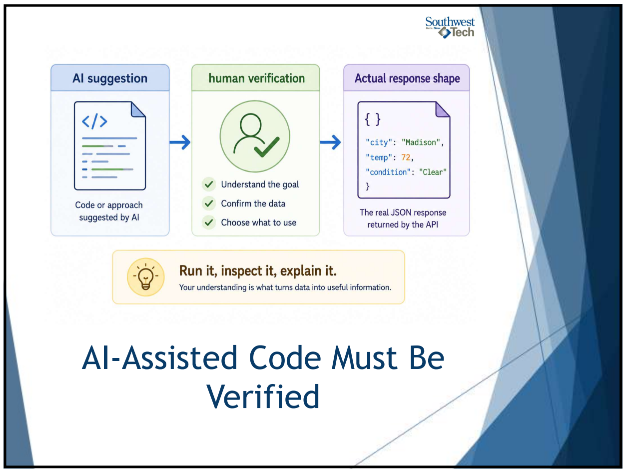


Southwest Tech

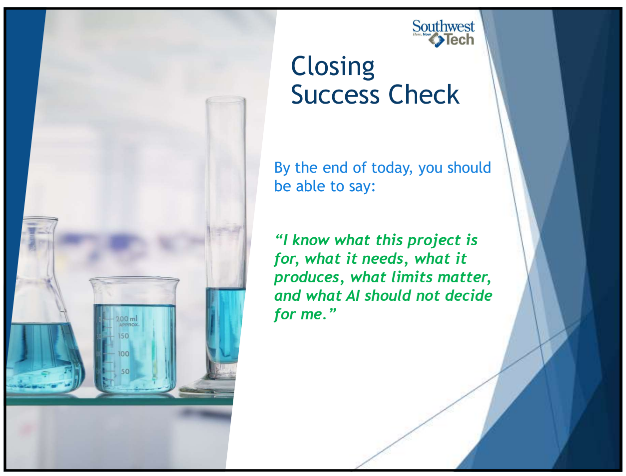
AI Use

- AI can suggest code that looks believable but does not match the actual response.
- You must inspect the response, run the code, and explain what human decision you made.

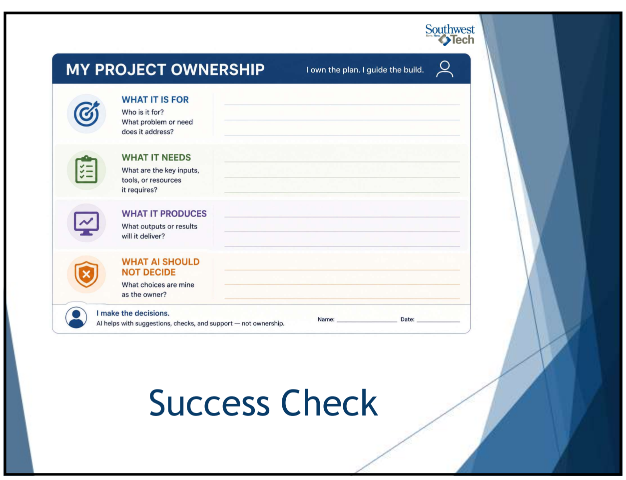
36


AI suggestion
Code or approach suggested by AI
human verification
Understand the goal
Confirm the data
Choose what to use
Actual response shape
{ }
"city": "Madison",
"temp": 72,
"condition": "Clear"
}
The real JSON response returned by the API
Run it, inspect it, explain it.
Your understanding is what turns data into useful information.
AI-Assisted Code Must Be Verified

37


Closing
Success Check
By the end of today, you should be able to say:
"I know what this project is for, what it needs, what it produces, what limits matter, and what AI should not decide for me."

38


MY PROJECT OWNERSHIP
I own the plan. I guide the build.
WHAT IT IS FOR
Who is it for? What problem or need does it address?
WHAT IT NEEDS
What are the key inputs, tools, or resources it requires?
WHAT IT PRODUCES
What outputs or results will it deliver?
WHAT AI SHOULD NOT DECIDE
What choices are mine as the owner?
I make the decisions.
AI helps with suggestions, checks, and support — not ownership.
Name:
Date:
Success Check

39

40